



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

Course Code: SE 200

Class: BSCS-13E

Software Engineering

Instructor: Sir Sarosh Tahir

Project Deliverable 3

Team Members

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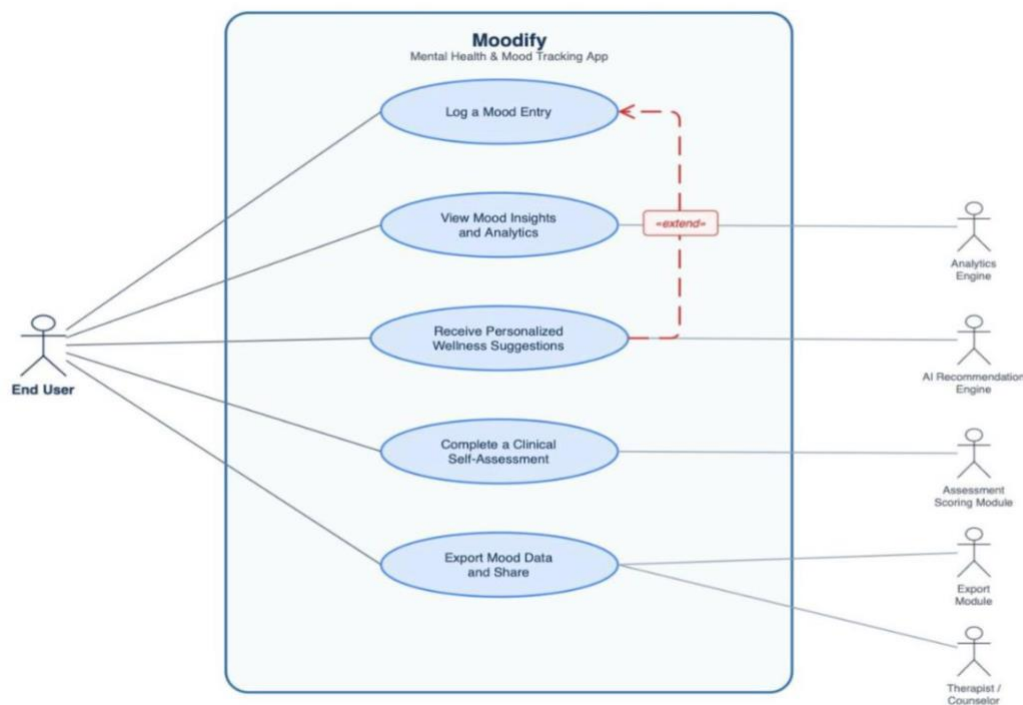
1. Use Case Diagram (Final)

Tool: Draw.io (XML)

The finalized use case diagram shows the Moodify system boundary with one primary actor (End User) on the left and five secondary actors on the right: Analytics Engine, AI Recommendation Engine, Assessment Scoring Module, Export Module, and Therapist/Counselor. Five use cases are shown inside the system boundary, corresponding directly to the five use case descriptions from Deliverable 2.

The diagram includes one extend relationship where Receive Personalized Wellness Suggestions extends Log a Mood Entry. This reflects the behavior described in UC-03, which states that suggestions are automatically triggered after saving a mood entry, making it an extension of the logging use case rather than an independent action.

Secondary actors are placed on the right side following standard UML convention, representing system-level components (engines and modules) rather than human users. The End User connects to all five use cases since they are the initiator of every interaction. The Therapist/Counselor actor is connected only to the Export Mood Data and Share use case, since that is the only flow where an external human recipient is involved.





2. Class Diagram

Tool: Draw.io (XML)

The class diagram models the static structure of the Moodify system with 12 classes organized into color-coded domain groups. Each class maps directly to one or more functional requirements to maintain clear traceability.

Color Group	Classes	Represents
Blue	User	Core identity and authentication (FR-01)
Green	Streak, Notification	Engagement and reminders (FR-12, FR-15)
Yellow	MoodEntry, EmotionTag, ActivityTag, GratitudeJournal	Logging and journaling (FR-02 to FR-04, FR-10)
Red	AnalyticsDashboard, WellnessSuggestion, BreathingExercise	AI and analytics features (FR-06, FR-07, FR-09)
Purple	ClinicalAssessment, ExportReport	Clinical and export features (FR-11, FR-13, FR-14)

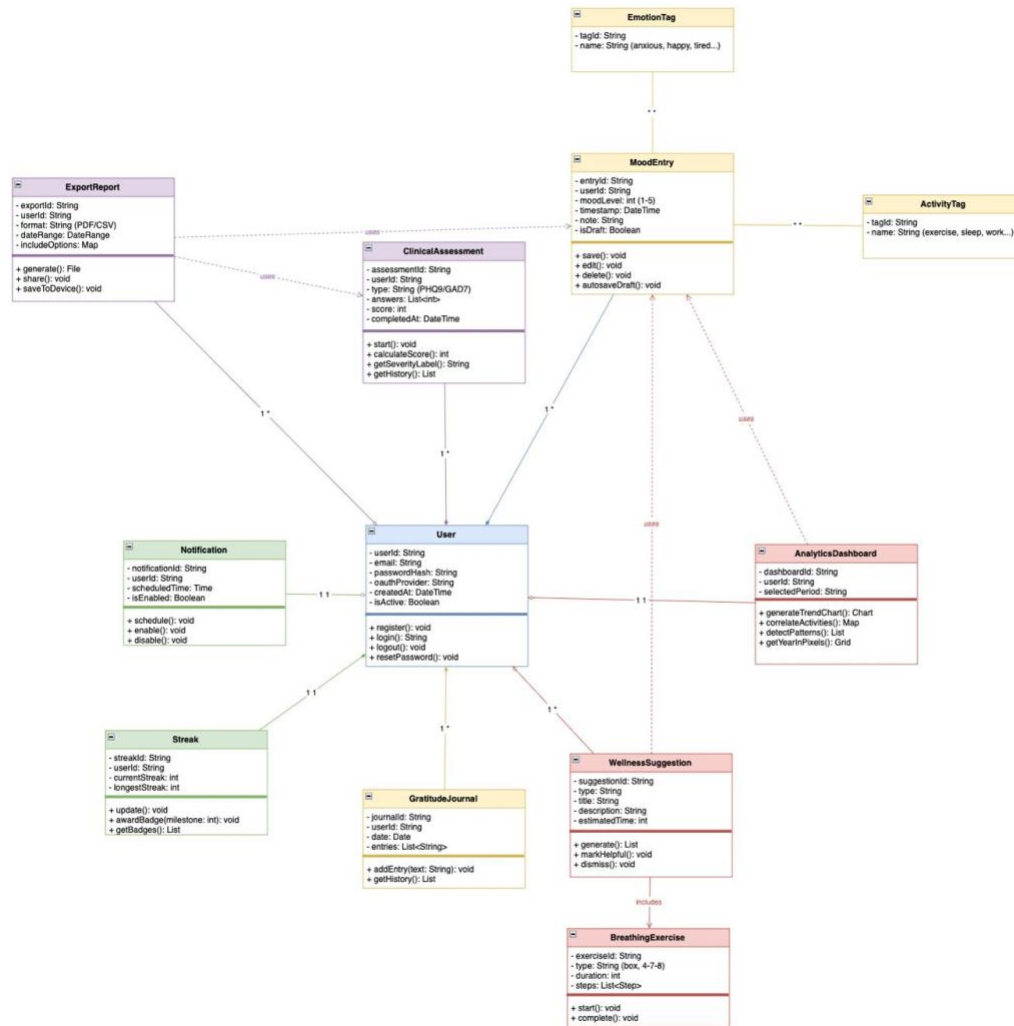
Relationships:

- Composition (filled diamond): User owns MoodEntry, Streak, ClinicalAssessment, and GratitudeJournal. These entities cannot exist without a User.
- Aggregation (open diamond): User has Notification, AnalyticsDashboard, WellnessSuggestion, and ExportReport. These have looser ownership.
- Many-to-many association: MoodEntry connects to both EmotionTag and ActivityTag, since each entry can have multiple tags and each tag can appear across entries.
- Dependency (uses): AnalyticsDashboard, WellnessSuggestion, and ExportReport all depend on MoodEntry data for their operations.
- Association: WellnessSuggestion includes BreathingExercise as one of its suggestion types.



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3. Sequence Diagram 1 — Log a Mood Entry (UC-01)

Tool: Draw.io (Mermaid)

Participants: User, MoodifyApp, MoodEntry, Streak

This sequence diagram captures the full interaction flow for logging a mood entry, which is the most frequently used feature in Moodify. It follows the 10-step main flow described in UC-01, starting from the user opening the app or tapping a reminder notification and ending with the home screen updating to reflect the logged mood.

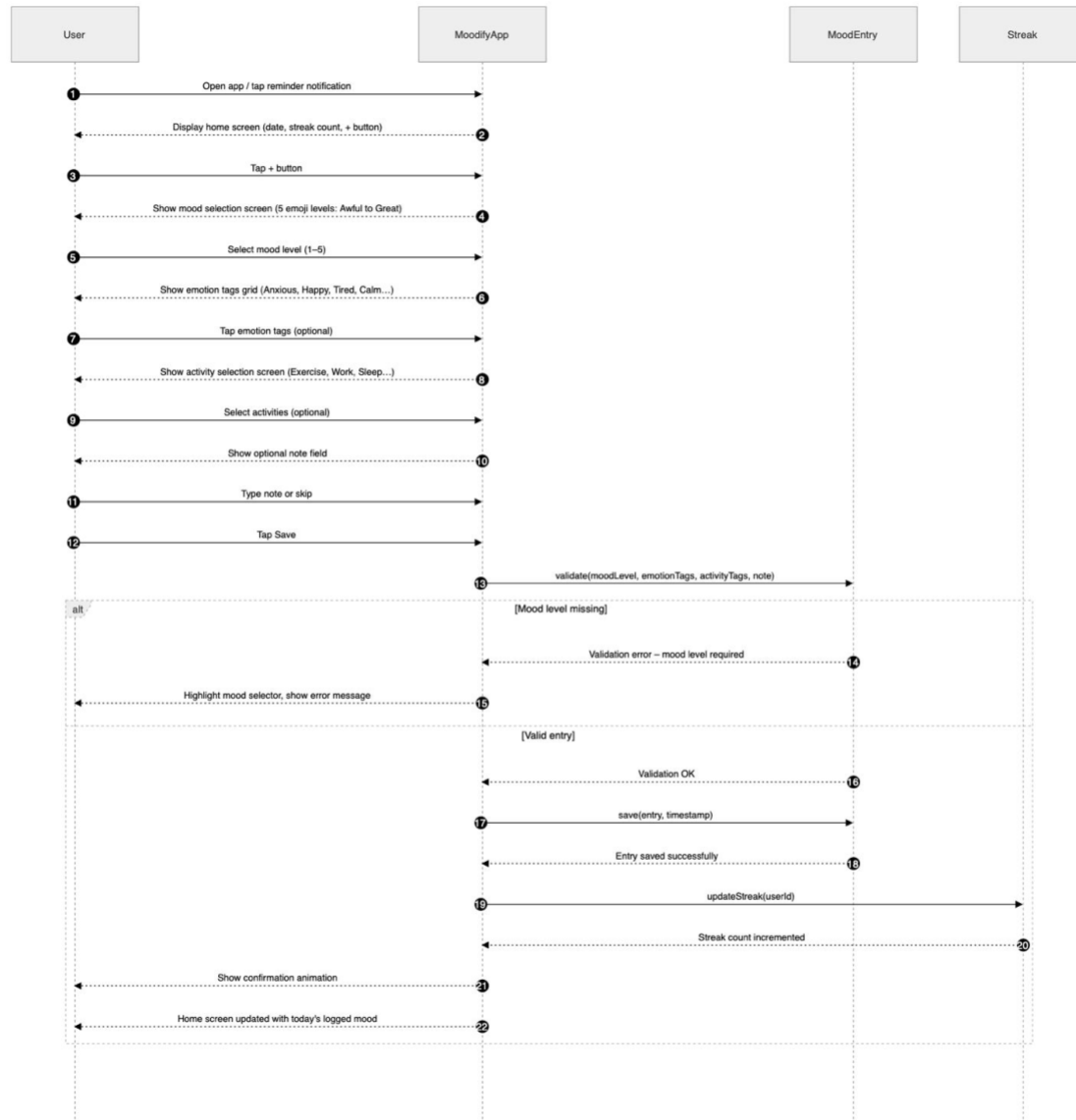
The diagram includes four participants. User and MoodifyApp handle the front-end interaction, while MoodEntry and Streak represent back-end system components. Streak is kept as a separate participant because it has distinct responsibilities under FR-12, including updating the daily counter and awarding milestone badges.

An alt block is used to represent the branching logic at the validation step. In the error path, the user attempts to save without selecting a mood level, which triggers E3: the system highlights the mood selector and shows an error message. In the happy path, the entry passes validation, gets saved with a timestamp, the streak counter is updated, and a confirmation animation is displayed. This approach keeps both paths visible in a single diagram rather than splitting them across two separate ones, which is standard practice for sequence diagrams.



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4. Sequence Diagram 2 — View Mood Insights and Analytics (UC-02)

Tool: Draw.io (Mermaid)

Participants: User, MoodifyApp, AnalyticsEngine

The second sequence diagram models the analytics viewing flow, which is the most data-intensive use case in Moodify. The AnalyticsEngine is kept as a separate participant because UC-02 explicitly lists it as a secondary actor with its own responsibilities including trend generation, activity correlation, and pattern detection.

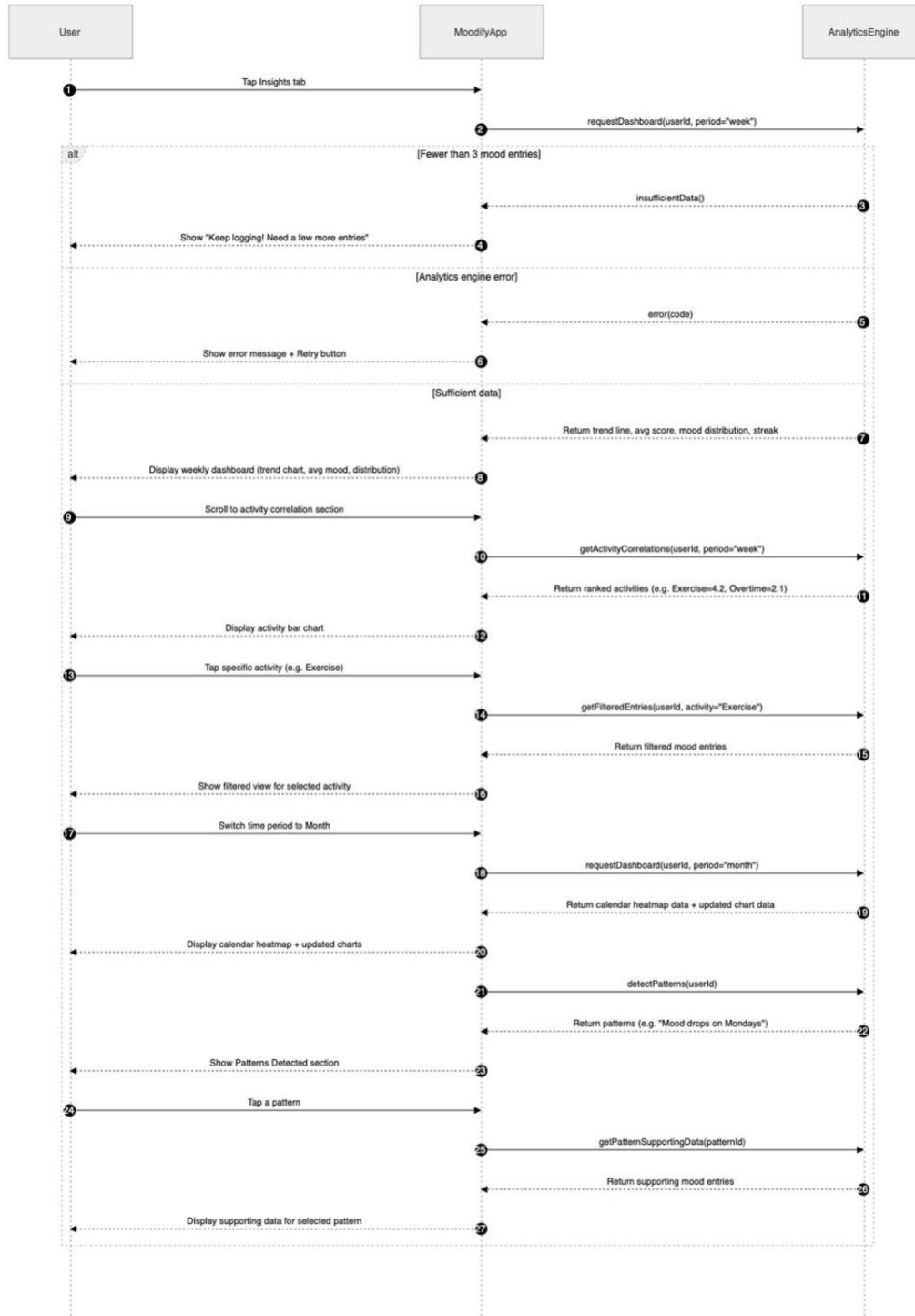
The diagram starts with three alt branches at the top to handle edge cases before proceeding to the main flow. The first branch covers insufficient data (fewer than 3 mood entries), where the system shows a friendly prompt encouraging the user to keep logging. The second branch handles analytics engine errors, displaying a retry button. The third branch is the happy path.

In the happy path, the flow progresses through several stages: loading the weekly dashboard with trend lines and mood distribution, displaying activity correlation bar charts that rank activities by their associated average mood, filtering entries by a specific activity when the user taps one, switching to the monthly view with a calendar heatmap, running AI pattern detection to surface observations like mood drops on Mondays, and finally allowing the user to drill into supporting data for any detected pattern.



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5. Activity Diagram — Log a Mood Entry (UC-01)

Tool: Draw.io

Based on: UC-01 through UC-05, FR-01 (authentication), FR-12 (streaks), and all exception flows

Unlike the sequence diagrams which each focus on a single use case, the activity diagram provides a unified control-flow view of the entire Moodify application. It captures the complete user journey from opening the app to exiting, covering all five use cases and their decision points in one diagram. This gives a holistic picture of how the different features connect and how the user navigates between them.

Authentication Gate:

The flow begins at a Start node and immediately moves to Open Moodify App, followed by a Check Authentication decision. If the user is not logged in, they are directed to the Login/Register Screen where they can authenticate via email or Google OAuth (FR-01). Once authenticated, or if they were already logged in, the system displays the Home Dashboard.

Action Fork — Four Parallel Branches:

From the Home Dashboard, the user reaches a fork node labeled User Selects Action with a decision diamond asking What action does the user choose? This branches into four parallel paths corresponding to the four main areas of the app:

Branch 1: Log Mood (UC-01)

The user taps the + Log Mood Button, selects a mood level on a 1 to 5 emoji scale, optionally tags emotions and activities, and adds an optional note (max 500 characters). After tapping Save, a Mood Level Selected? decision checks whether a mood was actually chosen. If not, an error message is shown asking the user to select a mood, and the flow loops back. If yes, the entry is saved locally and the streak is updated. The system then checks if there are 5 or more entries to trigger the AI Suggestion Engine, after which a confirmation screen with suggestion cards is displayed. This covers the main flow, alternative flows, and exception flow E3 from UC-01, as well as the extend relationship with UC-03.

Branch 2: View Insights (UC-02)

The user taps the Insights Tab, which triggers a Check 3 or more Entries? decision. If there are fewer than 3 entries, a Keep Logging prompt is shown encouraging the user to log more before analytics become available (UC-02 alternative flow A1). If sufficient data exists, the system renders mood trend charts across weekly, monthly, and yearly periods, shows activity correlations alongside AI-detected patterns, and allows the user to drill into pattern details.



Branch 3: Assessment (UC-04)

The user taps Assessments, which first checks Taken in Last 24 Hours? If yes, the system shows a Come Back Later message and lets the user view their last score (UC-04 exception flow E2). If not, the user is shown the PHQ-9 or GAD-7 intro screen with a disclaimer, then answers questions one by one. The system calculates the total score and severity level. A critical decision follows: Score Severe (20 or above, or Q9 flagged)? If yes, crisis resources are displayed and the user must acknowledge the screen before proceeding (UC-04 exception flow E1). If not, results with recommendations are shown and saved to history.

Branch 4: Export Data (UC-05)

The user taps Export/Share with Therapist, then selects a date range and format (PDF or CSV). A decision checks Any Data in Range? If no data exists for the selected period, an error message is shown (UC-05 exception flow E1). If data exists, the system generates the report file, shows a preview, and the user can share or save the file.

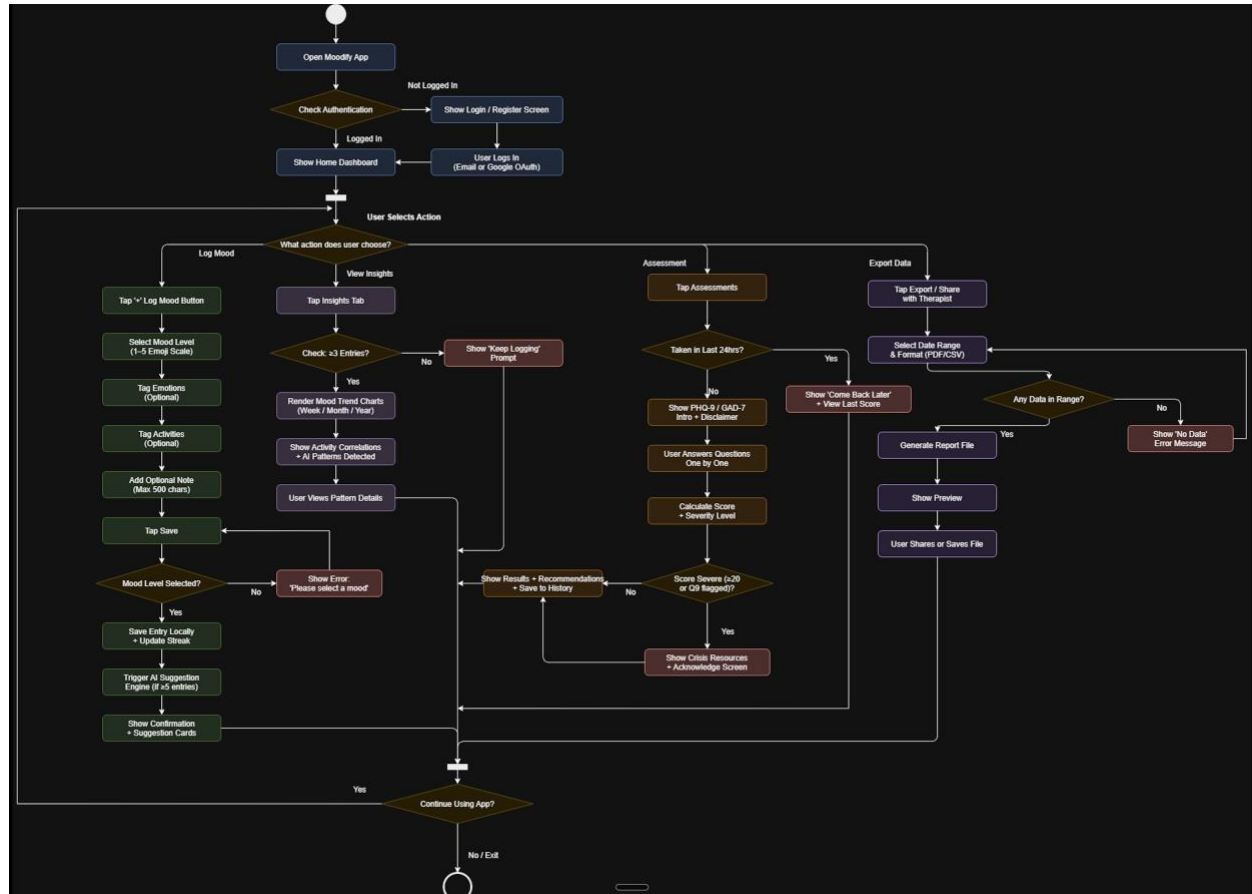
Convergence and Loop:

All four branches converge at a join node, leading to a final Continue Using App? decision. If the user wants to continue, the flow loops back to the action selection fork on the Home Dashboard. If the user chooses to exit, the flow reaches the End node. This loop structure reflects the real usage pattern where users typically perform multiple actions in a single session, for example logging a mood and then checking their insights.



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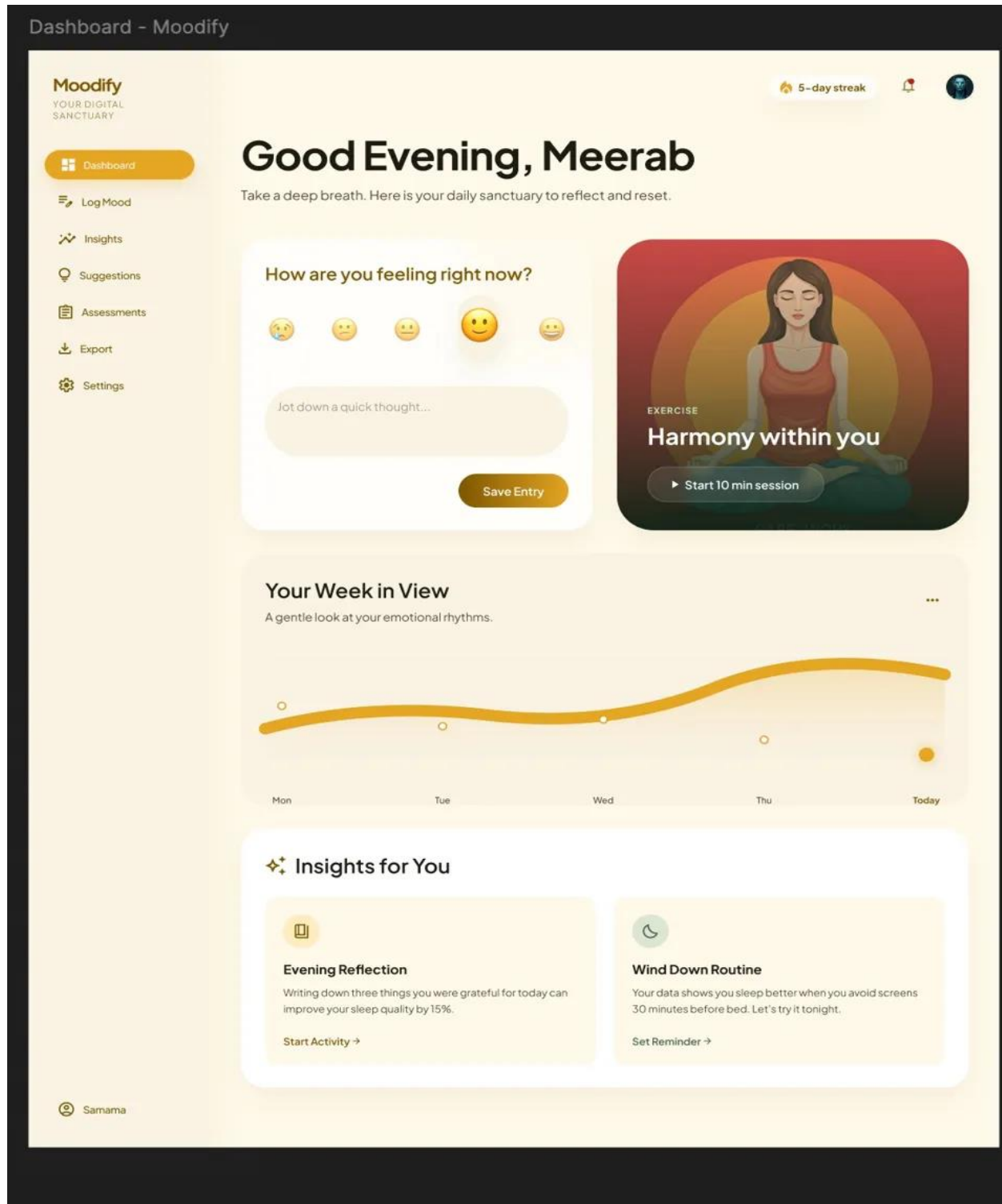
7. Figma Clickable Prototype

Tool: Figma

The Figma prototype was refined this week with updated screens that align with the finalized use case descriptions and functional requirements. The prototype now includes a polished dashboard screen, a Year in Pixels insights page, and working navigation between the main sections of the app. All screens use the Moodify design system with warm, earthy tones and a calming visual language that matches the mental health focus of the application.

Dashboard Screen:

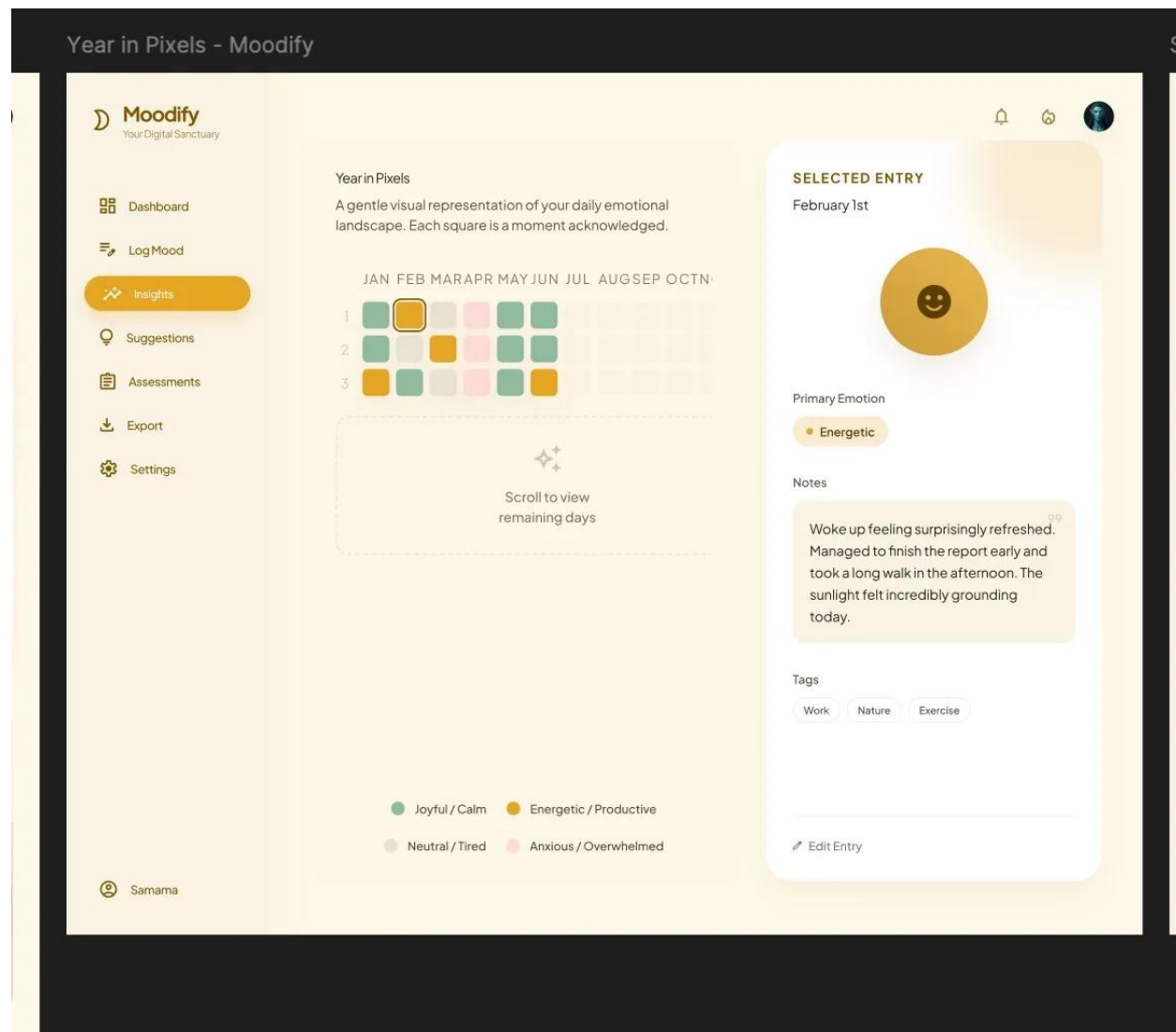
The dashboard serves as the main landing page after login. It displays a personalized greeting, a quick mood logging widget with emoji selectors and a note field, a breathing exercise card, a weekly mood trend chart, and an AI-powered Insights for You section with contextual suggestions. The 5-day streak badge is shown in the top-right corner alongside notification and profile icons.





Year in Pixels / Insights Screen:

The Insights screen features the Year in Pixels view, which displays a grid of colored squares representing the user's daily mood entries across the entire year. Each color corresponds to a mood category: green for Joyful/Calm, gold for Energetic/Productive, gray for Neutral/Tired, and pink for Anxious/Overwhelmed. Tapping a specific day opens a side panel showing the selected entry with the primary emotion, notes, and associated tags. The user can also edit the entry directly from this view.





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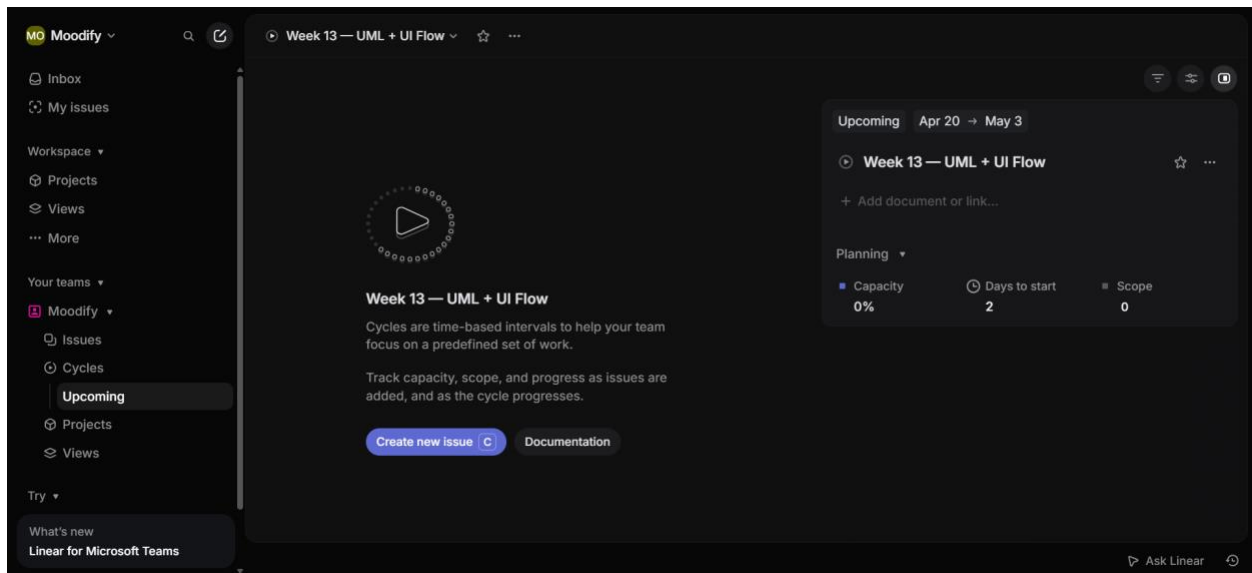
Figma Prototype Link:

<https://www.figma.com/proto/6vdpLrYCWUPflvdUOinY3x/Untitled?node-id=9-56&p=f&t=v337jNvcRAasiwI4-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1>



Linear Spring Update:

A new sprint cycle titled 'Week 13 — UML + UI Flow' was created in Linear for the period of Apr 20 to May 3. This cycle represents the current week's deliverables including UML diagrams, activity diagram, sequence diagrams, and Figma prototype work.



Six tasks were created, assigned, and completed by the team during this sprint.

Issue	Task	Assigned To	Status
MOO-24	Use Case Diagram — Draw.io	Ahmed Raza	Done
MOO-25	Class Diagram — Draw.io	Bilal Rana	Done
MOO-26	Sequence Diagram 1 — Draw.io	Bilal Rana	Done
MOO-27	Sequence Diagram 2 — Draw.io	Meerab Chaudry	Done
MOO-28	Activity Diagram — Draw.io	Meerab Chaudry	Done
MOO-29	Figma Clickable Prototype — Figma	M. Samama Usman	Done



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All tasks were marked as Done upon completion. Screenshots of each issue with full descriptions are included below.

Week 13 — UML + UI Flow

6 issues

Done 6

- MOO-29 Figma Clickable Prototype — Figma Apr 18
- MOO-28 Activity Diagram — Draw.io Apr 18
- MOO-27 Sequence Diagram 2 — Draw.io Apr 18
- MOO-26 Sequence Diagram 1 — Draw.io Apr 18
- MOO-25 Class Diagram — Draw.io Apr 18
- MOO-24 Use Case Diagram (Final) — Draw.io Apr 18

Upcoming Apr 20 → May 3

Week 13 — UML + UI Flow

+ Add document or link...

Planning

Capacity 15% Days to start 2 Scope 6

Assignees Labels Priority Projects

No assignee 6



Week 13 — UML + UI Flow > MOO-29 Figma Clickable Prototype — Figma ☆ ...

Figma Clickable Prototype — Figma

Designed a fully clickable Figma prototype with 11 screens including Login, Sign Up, Dashboard, Log Mood, Insights, Year in Pixels, Suggestions, PHQ-9 Assessment, Clinical Check-in, Export Data, Results, and Settings. All screens are connected with working navigation flow.



+ Add sub-issues

Activity

Unsubscribe



mchaudry.bs23seecs@seecs.edu.pk created the issue · 28min ago



Bilal assigned to Muhammad Samama Usman · 14min ago



▶ Week 13 — UML + UI Flow > MOO-28 Activity Diagram — Draw.io ☆ ...

Activity Diagram — Draw.io

Created the Activity Diagram using [Draw.io](#) covering the complete Moodify app process flow. Includes 4 parallel activity paths: Mood Logging, View Insights, Clinical Assessment, and Export Data, with all decision points, error flows, and alternative paths mapped out.



+ Add sub-issues

Activity

Unsubscribe



mchaudry.bs23seecs@seecs.edu.pk created the issue · 28min ago



Bilal assigned to mchaudry.bs23seecs@seecs.edu.pk · 14min ago

Leave a comment...





▶ Week 13 — UML + UI Flow > MOO-26 Sequence Diagram 1 — Draw.io ☆ ...

Sequence Diagram 1 — Draw.io

Developed Sequence Diagram 1 using [Draw.io](#) illustrating the Mood Logging Flow. Shows interactions between User, MoodifyApp, MoodEntry, and Streak including home screen display, mood level selection, emotion and activity tagging, optional note entry, validation with error handling for missing mood level, entry save with timestamp, streak counter update, and confirmation animation



+ Add sub-issues

Activity

Unsubscribe



 mchaudry.bs23seecs@seecs.edu.pk created the issue · 35min ago

 Bilal self-assigned the issue · 8min ago



▶ Week 13 — UML + UI Flow > MOO-27 Sequence Diagram 2 — Draw.io ☆ ...

Sequence Diagram 2 — Draw.io

Developed Sequence Diagram 2 using [Draw.io](#) illustrating the AI Personalized Suggestions Flow. Shows interactions between User, MoodifyApp, SuggestionController, MoodRepository, AIEngine, and ContentLibrary including cache check, pattern analysis, trigger identification, and suggestion ranking.



+ Add sub-issues

Activity

Unsubscribe



mchaudry.bs23seecs@seecs.edu.pk created the issue · 28min ago



Bilal assigned to mchaudry.bs23seecs@seecs.edu.pk · 1min ago



▶ Week 13 — UML + UI Flow > MOO-25 Class Diagram — Draw.io ☆ ...

Class Diagram — Draw.io

Created the Class Diagram for Moodify using [Draw.io](#) containing 11 classes with full attributes and methods: User (central class), MoodEntry, EmotionTag, ActivityTag, ClinicalAssessment, AnalyticsDashboard, WellnessSuggestion, BreathingExercise, GratitudeJournal, Streak, Notification, and ExportReport. Relationships include associations, dependencies (uses), and composition (includes) with proper multiplicities (1-1 and 1-many).



+ Add sub-issues

Activity

Unsubscribe



mchaudry.bs23seecs@seecs.edu.pk created the issue · 29min ago



Bilal self-assigned the issue · 9min ago



▶ Week 13 — UML + UI Flow > MOO-24 Use Case Diagram (Final) — Draw.io ☆ ...

Use Case Diagram (Final) — Draw.io

Finalized the Use Case Diagram for Moodify using [Draw.io](#). The diagram includes 5 core use cases: Log a Mood Entry, View Mood Insights and Analytics, Receive Personalized Wellness Suggestions, Complete a Clinical Self-Assessment, and Export Mood Data and Share. The End User connects to all use cases. Secondary actors include Analytics Engine, AI Recommendation Engine, Assessment Scoring Module, Export Module, and Therapist/Counselor. An «extend» relationship is shown from Suggestions to Insights to Log Mood, indicating conditional extensions.



+ Add sub-issues

Activity

Unsubscribe



 mchaudry.bsce23seecs@seecs.edu.pk created the issue · 32min ago

 Bilal assigned to araza.bsce23seecs@seecs.edu.pk · 8min ago

10. Summary

This week focused on translating the requirements and use case descriptions from Deliverable 2 into visual models. The use case diagram captures the system boundary and actor relationships. The class diagram models the static structure with 12 classes mapped to functional requirements. The two sequence diagrams cover the most important interaction flows (mood logging and analytics viewing) with full coverage of main, alternative, and exception paths. The activity diagram shows the control flow for mood logging with decision points and loops. The Figma prototype brings the design to life with interactive screens. All tasks were tracked in Linear and complete